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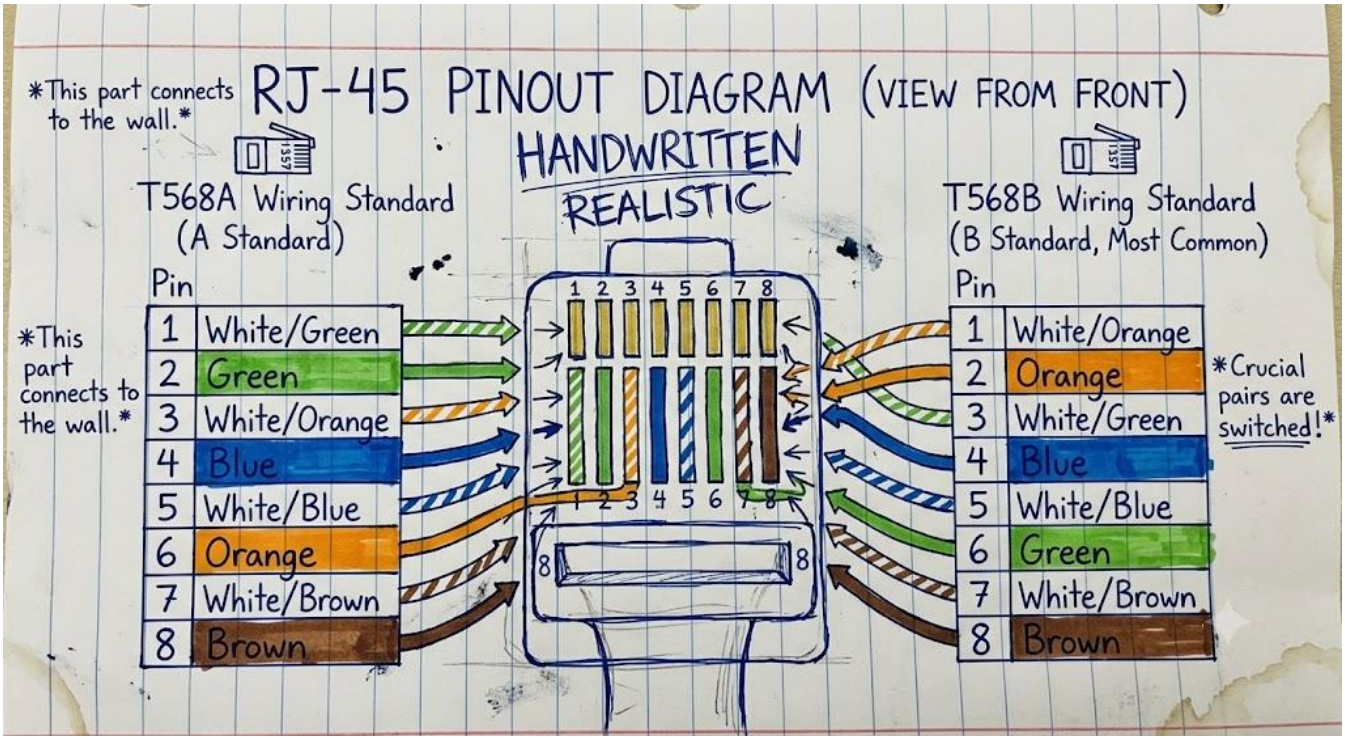
N+ - Network fundamentals and building Networks

Ethernet standard and connectos

TASKS:-1

Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

RESULT



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TASKS:-2

Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

RESULT

Ethernet Standard	Speed	Cable Type	Maximum Cable Length	Connector Type
10BASE-T	10 Mbps	Cat3 or better twisted-pair	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5 or better twisted-pair	100 m	RJ-45
1000BASE-T	1 Gbps	Cat5e or better twisted-pair	100 m	RJ-45

All three standards commonly use **RJ-45 connectors** and have a typical maximum copper cable segment length of **100 metres**. 1000BASE-T uses all four twisted pairs to achieve 1 Gbps.

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TASKS:-3

Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and whyThink about how transmit and receive pairs are wired in each cable type.

RESULT

If I need to connect **two laptops directly using Ethernet**, I would traditionally use an **Ethernet crossover cable**.

Reason:
A crossover cable swaps the **transmit (TX)** and **receive (RX)** pairs at one end. This allows the transmit signals from one laptop to reach the receive pins of the other laptop.

Crossover cable: .

Laptop ↔ Laptop — TX ↔ RX

Straight-through cable:

Laptop ↔ Switch/Router — TX → TX,

RX → RX

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TASKS:-4

List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

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RESULT

No.	Real-World Device	Ethernet Standard	Use
1	Wi-Fi Router	1000BASE-T	Connects to a computer or switch using a Gigabit Ethernet cable.
2	Desktop Computer	1000BASE-T	Provides a fast wired Internet/network connection.
3	Network Printer	100BASE-TX / 1000BASE-T	Connects to the college/home network for printing.